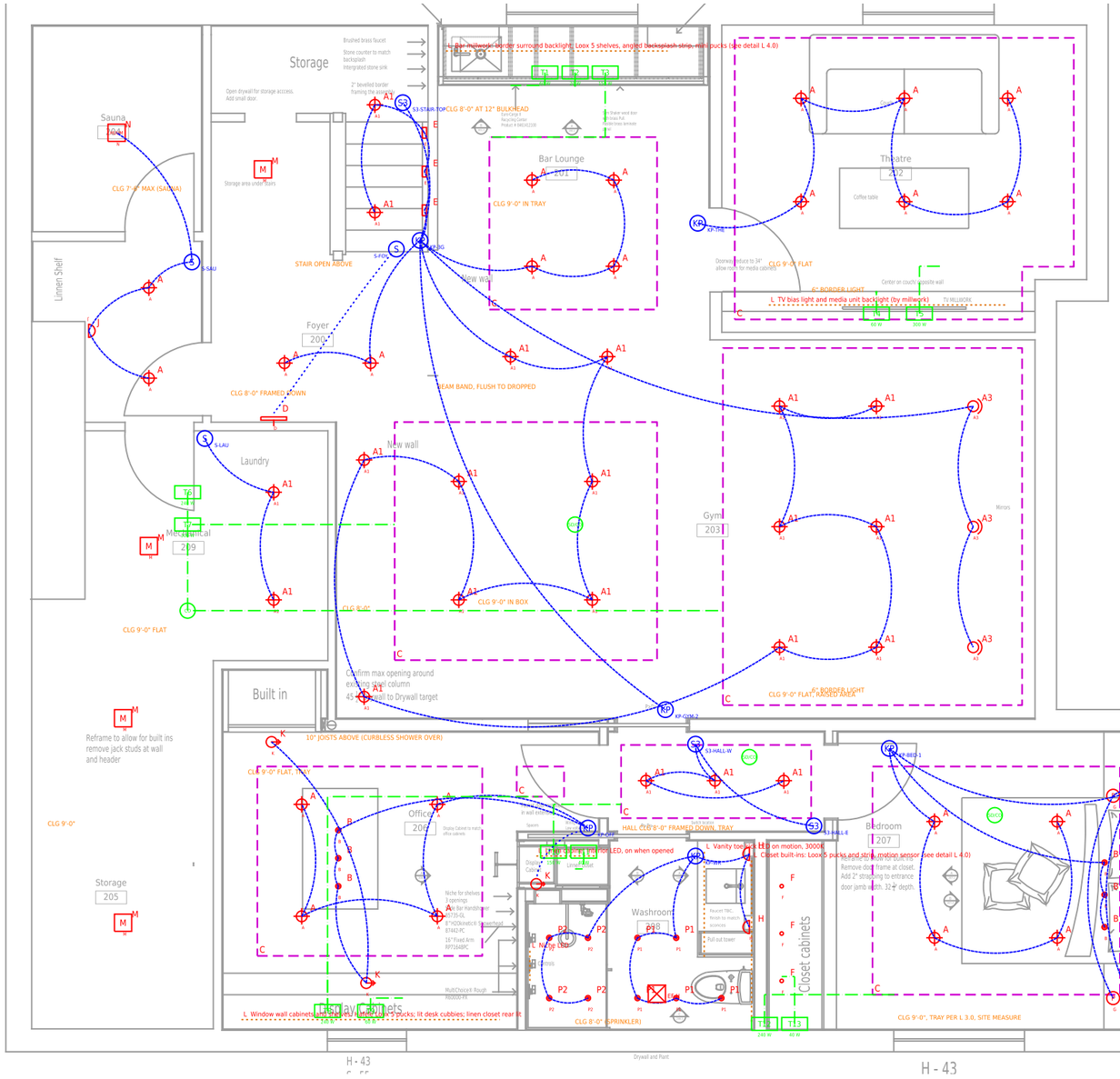


LOWER LEVEL LIGHTING AND SWITCHING PLAN Scale 3/16" = 1'-0"

by William

LEGEND

A	3" trimless recessed pot, 2700K
A1	3" trimless recessed pot, 3000K
A3	3" adjustable wall-wash pot, 3000K (mirror wall)
P1	1" micro pin, washroom
P2	1" micro pin, wet rated (shower)
B	1" micro pin, adjustable (ceiling gallery)
C	Linear LED in KLUS NISA-KRA trimless profile: lit tray, cove or border (see L 3.0)
D	Surface picture light, brass
E	Recessed wall step light
F	Hafele Loox 5 1" recessed puck (cabinets)
G	Bedside pendant, Huron PD95108, frosted acrylic tube 32-3/4" H, Bedroom 207
H	Vanity sconce, Remi 18", nickel
J	Wall sconce, sauna foyer
K	Art light, office
L	LED tape, millwork (toe kick, under shelf, backlight, bias)
M	LED surface fixture with motion sensor
N	Sauna light, by HUUM

KP	Smart keypad with scenes
S / S3	Switch / 3-way switch
SD/CO	Smoke and CO alarm, hardwired, interconnected
EF-H	Exhaust fan on humidity sensor
- - -	Control link, suggested grouping
T#	Low voltage driver (transformer), see L 2.1
-.-	Low voltage lead, class 2

NOTES

- Ceiling heights per Dean Thomas A3.2 and site; confirm on site before layout.
- Lighting runs as a standalone smart system for the lower level (Lutron, or to match the main floor).
- Switches and keypads 48" AFF to centre. Bedroom keypads built into headboard over night tables.
- Recessed fixtures in insulated or exterior ceilings to be IC rated and airtight.
- Shower fixtures wet rated; washroom fixtures damp rated.
- Cabinet lighting on door contact or motion (see L 4.0). Drivers and LV leads per L 2.1.
- Fixture types and products in schedule L 2.0. Items marked Client selection to be confirmed.

109 Rivers Bend Way, Dead Man's Flats AB
ByWilliam Design | 647.227.4825 | william@bywilliam.ca
Design intent only. Circuits, loads and panel by licensed electrician.

Fabi Residence - Lighting and Switching Plan
Rev 2, Oct 19, 2026
Plan - L 1.0

LIGHTING FIXTURE SCHEDULE

Type	Description	Product	CCT	Finish	Mounting	Control	Qty	Rooms
A	3" trimless recessed pot, 2700K	TBC, 3" trimless IC/AT LED	2700K	White trimless	Recessed, plaster-in trimless	Smart dimmer (Lutron)	22	Foyer 200, Bar Lounge 201, Theatre 202, Sauna Foyer 210, Office 206, Bedroom 207
A1	3" trimless recessed pot, 3000K	TBC, same family as A	3000K	White trimless	Recessed, plaster-in trimless	Smart dimmer (Lutron)	21	Stair, Gym 203, Mechanical 209 / Laundry, Hall
A3	3" adjustable wall-wash pot, 3000K (mirror wall)	TBC, wall-wash trim of A	3000K	White trimless	Recessed, 24" off wall	Smart dimmer (Lutron)	3	Gym 203
P1	1" micro pin, washroom	TBC, 1" low voltage pin, damp rated	3000K	White	Recessed, damp location	Smart dimmer, washroom	5	Washroom 208
P2	1" micro pin, wet rated (shower)	TBC, 1" low voltage pin, wet rated	3000K	White	Recessed, wet location	Smart dimmer, shower	4	Washroom 208
B	1" micro pin, adjustable (ceiling gallery)	TBC, 1" low voltage pin	2700K	White	Recessed in tray	Smart dimmer, art zone	6	Office 206, Bedroom 207
C	Linear LED in KLUS NISA-KRA trimless profile (cove or tray edge)	KLUS NISA-KRA (A18026) with KLUS LED tape	Per room	Trimless	Plaster-in, 1/2" drywall	Smart dimmer, cove zone	260 ft	Bar Lounge 201, Theatre 202, Gym 203, Office 206, Hall, Bedroom 207
D	Surface picture light, brass	Client selection	2700K	Brass	Wall, above art	Own switch (S-FOY)	1	Foyer 200
E	Recessed wall step light	TBC	3000K	White	Stair wall, about 8" above tread	Stair zone, 3-way	3	Stair
F	Hafele Loox 5 1" recessed puck (cabinets)	Hafele Loox 5 LED 3090, 1.2W	2700K	Per millwork	In millwork	Door contact or motion	by millwork	Bar Lounge 201, Office 206, Washroom 208, Bedroom 207
G	Bedside pendant, Huron PD95108, frosted acrylic tube 32-3/4" H, Bedroom 207	Huron PD95108-BG, 12W LED, 1100 lm (710 delivered), CRI 90, 120V, dry location	3000K (2700K by order, MOQ)	Brushed gold PD95108-BG	Ceiling, coaxial wire adjustable to 120" max, 38" min hang	Bed zone, 3-way, TRIAC/ELV dimming 100-10%	2	Bedroom 207
H	Vanity sconce, Remi 18", nickel	Remi 18" sconce, maker to confirm	3000K	Nickel	Wall, flanking mirror	Vanity zone	2	Washroom 208
J	Wall sconce, sauna foyer	Client selection	2700K	TBC	Wall	Sauna foyer zone	1	Sauna Foyer 210
K	Art light, office	Client selection (suggestions sent)	2700K	Brass	Per selection	Office art zone	3	Office 206
L	LED tape, millwork (toe kick, under shelf, backlight, bias)	By millwork	Per room	Hidden	In millwork	Per note	by millwork	Bar Lounge 201, Theatre 202, Washroom 208, Bedroom 207, Office 206
M	LED surface fixture with motion sensor	TBC, low profile LED	3500K	White	Surface, ceiling	Integral motion	4	Stair, Mechanical 209 / Laundry, Storage 205
N	Sauna light, by HUUM	HUUM lighting kit with Drop heater	Per HUUM	Per HUUM	Per HUUM	Switch outside sauna	1	Sauna 204

CONTROLS

Tag	Type	Controls
KP-3G	Keypad	3-gang at bottom of stairs: foyer and stair (3-way with S3-STAIR-TOP at top of stairs), wet bar (bar scenes), gym (3-way with hall doorway)
S3-STAIR-TOP	3-way	3-way at top of stairs (main floor landing): foyer and stair lights, paired with the KP-3G keypad at the bottom of the stairs
KP-THE	Keypad	Theatre pots, cove, bias; scenes Movie, Watch, Bright, Off
KP-GYM-2	Keypad	Gym, 3-way at doorway to hall and washroom
S-SAU	Switch	Sauna light (outside sauna) and sauna foyer
S-LAU	Switch	Laundry, at the entry door
KP-OFF	Keypad	Office pots, tray, art; latch side of office door
S3-HALL-W	3-way	Hall, 3-way at gym pocket door
S3-HALL-E	3-way	Hall, 3-way at bedroom door
KP-WR	Keypad	Washroom pots, shower, vanity; fan on humidity sensor
KP-BED-1	Keypad	Bedroom pots, tray, art, pendants; 3-way with bed
KP-BED-2	Keypad	Built into headboard over night table: pendants, room; 3-way with door
KP-BED-3	Keypad	Built into headboard over night table: pendants, room; 3-way with door
S-FOY	Switch	Foyer picture light (D) on its own switch, beside KP-3G at the bottom of the stairs

GENERAL NOTES

- All LED 90+ CRI. Match dimmer to driver (ELV, TRIAC or 0-10V).
- KLUS NISA-KRA trimless profile in all trays and borders, plastered into 1/2" gypsum (see L 3.0).
- LED drivers remote, in accessible locations: mechanical room, cabinets or access panels. Record locations.
- Cabinet lighting Hafele Loox 5 (see L 4.0).
- Smoke and CO alarms per NBC 2023 Alberta Edition: each bedroom, hall serving bedrooms, each storey; CO near fuel-burning appliances.
- Electrical permit by the electrician.

109 Rivers Bend Way, Dead Man's Flats AB
ByWilliam Design | 647.227.4825 | william@bywilliam.ca
Design intent only. Circuits, loads and panel by licensed electrician.

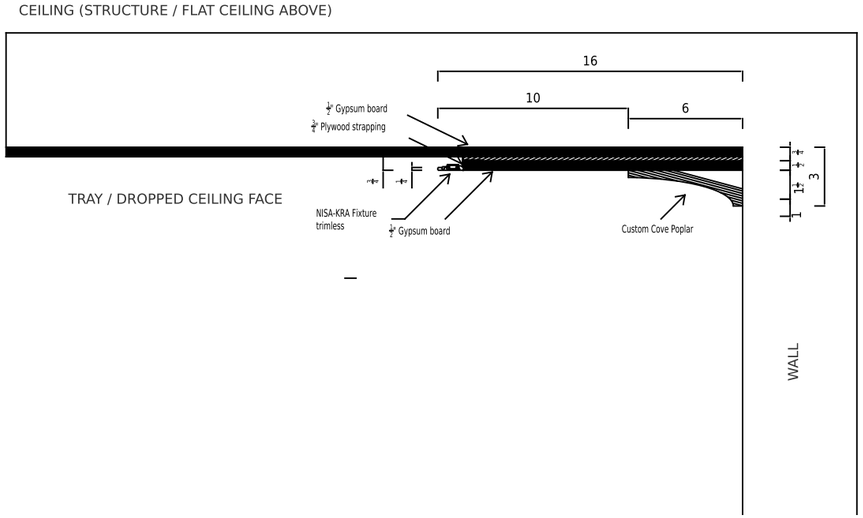
LOW VOLTAGE DRIVERS (TRANSFORMERS)

Tag	Room	Location	Serves	Load	Driver	Type	Load basis
T1	Bar Lounge 201	Bar base cabinet, end bay behind access panel	BAR-KICK	44 W	60 W	QTL Q-Tran QTM-eLED+WD (required for Kicker-Flat warranty)	BAR-KICK: QTL Kicker-Flat 2.0 W/ft x about 22 ft of border
T2	Bar Lounge 201	Bar base cabinet, end bay	BAR-PUCK	14 W	20 W	Hafele Loox5 24V dimmable	BAR-PUCK: Loox5 3090: 6 shelf pucks + 6 mini pucks in border top x 1.2 W
T3	Bar Lounge 201	Bar base cabinet, end bay	BAR-COVE	116 W	150 W	24V constant voltage, Lutron compatible	BAR-COVE cove 26 ft x 4.39 W/ft
T4	Theatre 202	Media unit, rear cabinet behind TV (PS80 access)	THE-TV	36 W	60 W	Hafele Loox5 24V dimmable	THE-TV: Loox5 3150 about 18 ft media unit backlight and TV bias, 2 W/ft
T5	Theatre 202	Media unit, rear cabinet behind TV	THE-COVE	211 W	300 W	24V constant voltage, Lutron compatible	THE-COVE cove 48 ft x 4.39 W/ft
T6	Mechanical 209 / Laundry	Driver panel on mechanical room wall	GYM-TRAY-B	170 W	240 W	24V constant voltage, Lutron compatible	GYM-TRAY-B cove 39 ft x 4.39 W/ft
T7	Mechanical 209 / Laundry	Driver panel on mechanical room wall	GYM-TRAY-C	223 W	300 W	24V constant voltage, Lutron compatible	GYM-TRAY-C cove 51 ft x 4.39 W/ft
T8	Office 206	Window wall cabinet base, accessible kick or bay	OFF-COVE, OFF-COVE2	167 W	240 W	24V constant voltage, Lutron compatible	OFF-COVE cove 32 ft x 4.39 W/ft; OFF-COVE2 cove 6 ft x 4.39 W/ft
T9	Office 206	Window wall cabinet base	OFF-MW	38 W	60 W	Hafele Loox5 24V dimmable	OFF-MW: Loox5: 12 shelf pucks, cubby strips about 10 ft, 3 pucks linen rear
T10	Washroom 208	Linen cabinet, top shelf behind panel (outside wet zone)	HALL-COVE	89 W	150 W	24V constant voltage, Lutron compatible	HALL-COVE cove 20 ft x 4.39 W/ft
T11	Washroom 208	Linen cabinet, top shelf behind panel	WR-MW	26 W	40 W	Hafele Loox5 24V dimmable	WR-MW: Loox5: linen 5 + display 3 pucks, niche 4 ft, toe kick 4 ft strip
T12	Bedroom 207	Closet built-in base drawer zone	BED-COVE	162 W	240 W	24V constant voltage, Lutron compatible	BED-COVE cove 37 ft x 4.39 W/ft
T13	Bedroom 207	Closet built-in base drawer zone	BED-BUILTIN	17 W	40 W	Hafele Loox5 24V dimmable	BED-BUILTIN: Loox5: 6 shelf pucks, 2 hanging-section strips about 5 ft

NOTES

- All drivers 24 VDC constant voltage, class 2, in accessible locations. Never inside a wall or above drywall without an access panel.
- Driver size = load / 0.8, next standard size. Coves from run length x 4.39 W/ft (KLUS tape 1440/1); millwork loads estimated, confirm on shop drawings.
- Dimming: cove drivers to suit the Lutron system (ELV / 0-10V). Hafele drivers dim through the Loox5 distributor and sensors. QTL Kicker-Flat needs a Q-Tran supply.
- LV leads: 18/2 for runs to 15 ft; 16/2 or 14/2 beyond. Keep voltage drop under 3%. Feed runs over 30 ft (theatre, gym C) from the centre or both ends.
- Run leads in the low voltage chases shown on 207 and 208, through millwork grommets, never loose above the trays.
- 120V feed to each driver location by the electrician. Record driver locations on the as-built.

TRIMLESS MUD-IN LIGHT KLUS NISA-KRA (A18026)



APPLICATION 1 COVE TRAY WITH CUSTOM POPLAR COVE

Full size detail as drawn by ByWilliam. Tray step 16": 10" flat return plus 6" poplar cove.

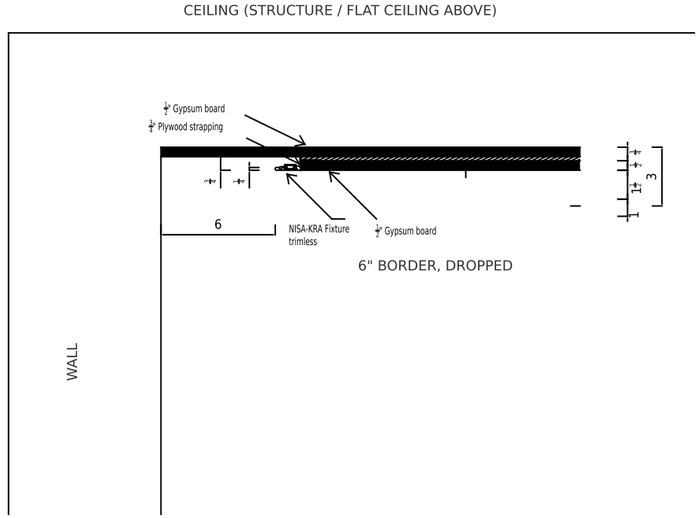
WHERE USED

Application 1: Bedroom 207 tray, Office 206 tray (from upper cabinets), Hall tray, Bar Lounge 201 tray, Gym coffer.

Application 2: Theatre 202 border, Gym 203 raised area border.

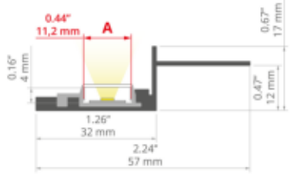
PROFILE

- KLUS NISA-KRA trimless fixture, ref. A18026. Aluminium, 2.2" wide x 0.7" high (57 x 17 mm).
- For 1/2" (12.5 mm) gypsum board. Lit edge (glow width) 0.44" (11.2 mm).
- One LED tape up to 0.31" wide, max 7.3 W/ft. IP20, dry locations only.
- Cover: KLUS LIGER-11 satin or frosted (confirm with rep). Cut to length by KLUS where runs exceed stock lengths.
- Joins with connectors into continuous runs; wire out through end cap or back of extrusion.



APPLICATION 2 6" BORDER LIGHT

Full size detail as drawn by ByWilliam. 6" border with NISA-KRA at the drop edge.



KLUS NISA-KRA
manufacturer section

INSTALL SEQUENCE

- Frame the tray or border and 3/4" plywood strapping as detailed so the profile has solid backing.
- Fix the profile to the strapping with the KLUS mounting track and screws before board goes on. Check level and straight; the lit line shows every wave.
- Run LED feed and drop to the driver location before board. Leave an access point at each driver.
- Hang 1/2" gypsum tight to the profile flange. Tape and mud over the flange like a corner bead, three coats, feathered. Keep compound off the LED channel.
- Prime and paint. Install LED tape and clip in the cover last, after paint and sanding dust are gone.

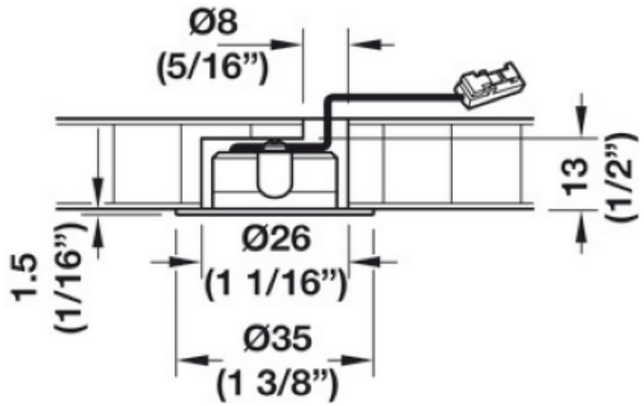
POWER AND CONTROL

- 24V constant voltage LED tape, 90+ CRI. 2700K in bar, theatre, office, bedroom; 3000K in gym and hall.
- Remote drivers in accessible locations (mechanical room, cabinets or access panels). Size drivers to 80% load.
- Dimming: driver to match the smart system (Lutron ELV or 0-10V). Confirm with the electrician.
- Feed long runs from both ends or the centre to avoid voltage drop.
- Rep: KLUS, amagency.com. Tape, output and driver to be confirmed before order.

109 Rivers Bend Way, Dead Man's Flats AB
ByWilliam Design | 647.227.4825 | william@bywilliam.ca
Design intent only. Circuits, loads and panel by licensed electrician.

Fabi Residence - Mud-in Trimless Lighting Detail
Rev 2, Oct 19, 2026
Detail - L 3.0

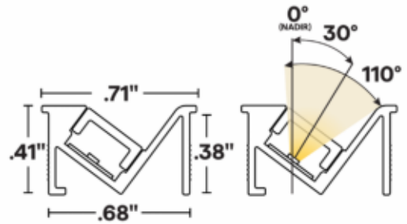
by William



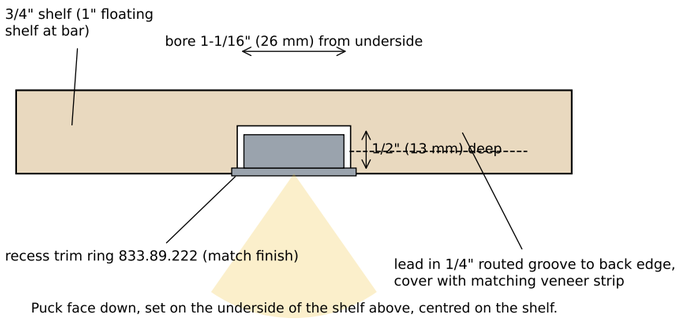
Hafele Loox5 LED 3090, manufacturer drawing

LOOX5 LED 3150 STRIP

- 8 mm tape, 6.5 W/m, 945-1080 lm/m, CRI 92.
- 2700K 833.76.265 / 3000K 833.76.266.
- Cut every 50 mm, max run 13.5 m.
- Set in recessed aluminium profile with frosted diffuser.
- Lead and connectors ordered separately.



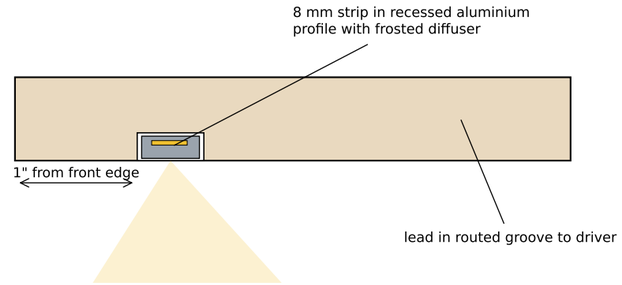
QTL Micro 5 Kicker-Flat, manufacturer drawing (detail 3 below)



Puck face down, set on the underside of the shelf above, centred on the shelf.

1 RECESSED PUCK IN SHELF HAFELE LOOX5 LED 3090

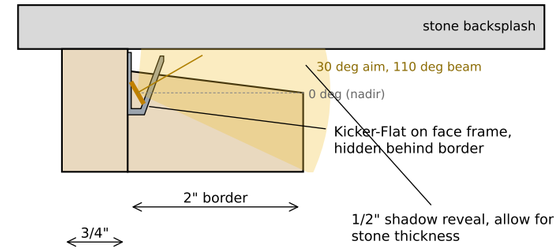
Section, half full size. Closet 207, linen and display 208, office 206, bar 201 shelves.



Light aimed back toward the wall so the source is not seen from the room.

2 UNDER-SHELF STRIP HAFELE LOOX5 LED 3150 IN PROFILE

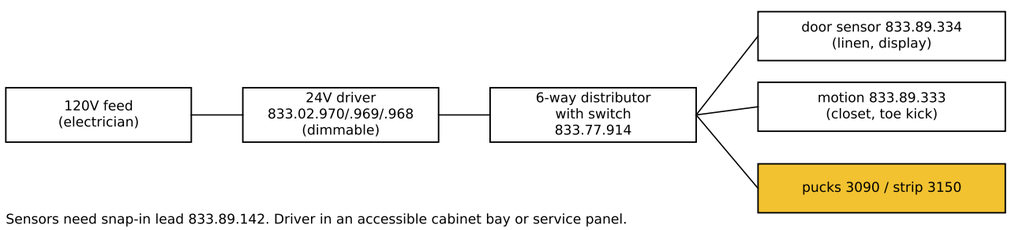
Section, half full size. Office shelves and cubbies, vanity toe kick.



Source never visible; light grazes the stone face inside the border.

3 WET BAR BORDER BACKLIGHT QTL MICRO 5 KICKER-FLAT

Plan section at the 2" bevelled border, full size. Bar Lounge 201, as drawn 201.3 (3 Sep).



Sensors need snap-in lead 833.89.142. Driver in an accessible cabinet bay or service panel.

4 LOOX5 CABINET CONTROL

Typical wiring, 24 V. One driver per cabinet group.

PRODUCTS

- Hafele Loox5 LED 3090 puck, 1.2 W, 75-85 lm, CRI >90, 2700K 833.75.170 / 3000K 833.75.171. 26 mm (1-1/16") bore, 13 mm deep, IP44 recessed.
- Hafele Loox5 LED 3150 strip, 8 mm, 6.5 W/m, CRI 92, 2700K 833.76.265 / 3000K 833.76.266. Cut every 50 mm. In aluminium profile with diffuser.
- QTL MICRO 5 KICKER-FLAT (KCKR-01), 24 VDC, 2.0 W/ft, 124 lm/ft at 3000K, CRI 94-98, warm dim 2000/3000 option. Q-Tran supply for warranty.
- Drivers: Hafele 24V dimmable 833.02.970 (20 W), .969 (40 W) or .968 (60 W). Size to 80% of load.

WHERE AND HOW SWITCHED

- Bar 201: Loox5 puck under each brass-edged shelf, 3 per side; border backlight on the bar button of the 3-gang at the stair.
- Washroom 208: Loox5 1" puck per shelf in linen and display cabinet, door sensor. Niche: LED at rear panel or each shelf. Vanity toe kick strip on motion.
- Office 206: pucks in shelving, lit desk cubbies, linen closet rear half lit.
- Bedroom 207 closet: Loox5 puck per shelf centre; hanging sections on motion.
- Low voltage wiring chases as drawn on 208 and 207 sheets.

109 Rivers Bend Way, Dead Man's Flats AB
ByWilliam Design | 647.227.4825 | william@bywilliam.ca
Design intent only. Circuits, loads and panel by licensed electrician.

Fabi Residence - Cabinet Lighting Details
Rev 2, Oct 19, 2026
Detail - L 4.0